

CBSE Class 10 Mathematics — Areas Related to Circles — Solutions

1. 1. If the circumference of a circle is 132 cm, then its radius is:

Answer: (B)

Solution:

1. Recall the formula for circumference $C = 2\pi r$.
2. Substitute the given value $132 = 2 \times \frac{22}{7} \times r$.
3. Solve for r .

Derivation:

$$\begin{aligned}C &= 2\pi r \\ \Rightarrow 132 &= 2 \times \frac{22}{7} \times r \\ \Rightarrow 132 &= \frac{44}{7}r \\ \Rightarrow r &= \frac{132 \times 7}{44} \\ \Rightarrow r &= 3 \times 7 = 21 \text{ cm}\end{aligned}$$

Areas Related to Circles — Circumference of a circle

2. 2. The area of a circle whose diameter is 14 cm is:

Answer: (C)

Solution:

1. Find the radius $r = \frac{d}{2} = \frac{14}{2} = 7$ cm.
2. Use the area formula $A = \pi r^2$.
3. Calculate $A = \frac{22}{7} \times 7^2$.

Derivation:

$$\begin{aligned}r &= 7 \text{ cm} \\ \Rightarrow A &= \frac{22}{7} \times 7 \times 7 = 22 \times 7 = 154 \text{ cm}^2\end{aligned}$$

Areas Related to Circles — Area of a circle

3. 3. The area of a quadrant of a circle with radius $r = 7$ cm is:

Answer: (A)

Solution:

1. A quadrant is $\frac{1}{4}$ of a circle.
2. Calculate total area $A = \pi r^2 = \frac{22}{7} \times 7^2 = 154 \text{ cm}^2$.
3. Divide by 4.

Derivation:

$$\text{Area} = \frac{1}{4}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 7 \times 7 = \frac{154}{4} = 38.5 \text{ cm}^2$$

Areas Related to Circles — Area of a sector

4. **4.** If the perimeter of a semi-circular protractor is 36 cm, find its diameter.

Answer: (B)

Solution:

1. The perimeter of a semicircle is $\pi r + 2r$.
2. Set $\pi r + 2r = 36$.
3. Solve for r and then find $d = 2r$.

Derivation:

$$\begin{aligned} r(\pi + 2) &= 36 \\ \Rightarrow r\left(\frac{22}{7} + 2\right) &= 36 \\ \Rightarrow r\left(\frac{36}{7}\right) &= 36 \\ \Rightarrow r &= 7 \\ \Rightarrow d = 2r &= 14 \text{ cm} \end{aligned}$$

Areas Related to Circles — Perimeter of a semicircle

5. **5.** The area of a sector of a circle with radius 6 cm and central angle 60° is:

Answer: (D)

Solution:

1. Use the formula $\text{Area} = \frac{\theta}{360} \times \pi r^2$.
2. Substitute $\theta = 60$ and $r = 6$.
3. Calculate the result.

Derivation:

$$\text{Area} = \frac{60}{360} \times \pi \times 6^2 = \frac{1}{6} \times \pi \times 36 = 6\pi \text{ cm}^2$$

Areas Related to Circles — Area of a sector

6. **6.** Assertion (A): The area of a circle with radius 7 cm is 154 cm^2 . Reason (R): The area of a circle is given by the formula πr^2 , where r is the radius.

Answer: (A)

Solution:

1. Recall formula πr^2 .
2. Substitute $r = 7$ and $\pi = 22/7$.
3. Calculate $22/7 \times 7 \times 7 = 154$.

Derivation:

$$Area = \pi r^2$$

$$Area = \frac{22}{7} \times 7 \times 7$$

$$Area = 22 \times 7 = 154 \text{ cm}^2$$

Areas Related to Circles — Area of a Circle

7. **7.** Assertion (A): The circumference of a circle with diameter 14 cm is 44 cm. Reason (R): The circumference of a circle is given by πd , where d is the diameter.

Answer: (A)

Solution:

1. Recall formula $C = \pi d$.
2. Substitute $\pi = 22/7$ and $d = 14$.
3. Calculate $22/7 \times 14 = 44$.

Derivation:

$$C = \pi d$$

$$C = \frac{22}{7} \times 14$$

$$C = 22 \times 2 = 44 \text{ cm}$$

Areas Related to Circles — Circumference of a Circle

8. **8.** Assertion (A): The area of a semicircle with radius r is $\frac{1}{2}\pi r^2$. Reason (R): A semicircle is half of a full circle.

Answer: (A)

Solution:

1. Area of full circle is πr^2 .
2. Semicircle is exactly half of a circle.
3. Divide the area formula by 2.

Derivation:

$$\text{Area of semicircle} = \frac{1}{2} \times \text{Area of circle}$$

$$\text{Area of semicircle} = \frac{1}{2}\pi r^2$$

Areas Related to Circles — Area of a Semicircle

9. **9.** Assertion (A): The length of an arc of a circle with radius r and central angle θ is $\frac{\theta}{360} \times 2\pi r$. Reason (R): The circumference of a circle is $2\pi r$.

Answer: (A)

Solution:

1. Arc length is a fraction of the circumference.
2. The fraction is determined by the central angle θ over 360° .
3. Combine to get $\frac{\theta}{360} \times 2\pi r$.

Derivation:

$$\text{Arc Length} = \frac{\theta}{360^\circ} \times 2\pi r$$

Areas Related to Circles — Length of an Arc

10. **10.** Assertion (A): The area of a quadrant of a circle with radius r is $\frac{1}{4}\pi r^2$. Reason (R): A quadrant is a sector with a central angle of 45° .

Answer: (C)

Solution:

1. The area of a quadrant is indeed $\frac{1}{4}\pi r^2$.
2. A quadrant implies a central angle of 90° , not 45° .
3. Therefore, Reason is false.

Derivation:

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

For quadrant, $\theta = 90^\circ$

$$\text{Area} = \frac{90}{360} \times \pi r^2 = \frac{1}{4}\pi r^2$$

Areas Related to Circles — Area of a Sector

11. **11.** Find the area of a sector of a circle with radius 6 cm if the angle of the sector is 60° .

Answer:

$$\frac{132}{7}\text{cm}^2$$

Solution:

1. Use the formula for the area of a sector $A = \frac{\theta}{360^\circ} \times \pi r^2$.
2. Substitute $\theta = 60^\circ$, $r = 6\text{ cm}$, and $\pi = \frac{22}{7}$ to calculate the area.

Derivation:

$$A = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 6^2$$

$$\Rightarrow A = \frac{1}{6} \times \frac{22}{7} \times 36$$

$$\Rightarrow A = 6 \times \frac{22}{7} = \frac{132}{7}\text{ cm}^2$$

Areas Related to Circles — Area of a Sector

12. **12.** The length of the minute hand of a clock is 14 cm . Find the area swept by the minute hand in 5 minutes.

Answer:

$$\frac{154}{3}\text{cm}^2$$

Solution:

1. Calculate the angle swept by the minute hand in 5 minutes: since it sweeps 360° in 60 minutes, in 5 minutes it sweeps $\frac{360}{60} \times 5 = 30^\circ$.
2. Apply the area formula for a sector with $r = 14 \text{ cm}$ and $\theta = 30^\circ$.

Derivation:

$$\begin{aligned}\theta &= \frac{360^\circ}{60} \times 5 = 30^\circ \\ \Rightarrow \text{Area} &= \frac{30^\circ}{360^\circ} \times \frac{22}{7} \times 14^2 \\ \Rightarrow \text{Area} &= \frac{1}{12} \times \frac{22}{7} \times 196 = \frac{1}{12} \times 22 \times 28 = \frac{616}{12} = \frac{154}{3} \text{ cm}^2\end{aligned}$$

Areas Related to Circles — Area of a Sector

- 13. 13.** If the perimeter and the area of a circle are numerically equal, then find the radius of the circle.

Answer:

2units

Solution:

1. Set the formula for circumference $2\pi r$ equal to the formula for area πr^2 .
2. Solve the equation $2\pi r = \pi r^2$ for r .

Derivation:

$$\begin{aligned}2\pi r &= \pi r^2 \\ \Rightarrow 2 &= r \text{ (since } r \neq 0\text{)}\end{aligned}$$

Areas Related to Circles — Area and Perimeter of a Circle

- 14. 14.** Find the area of a quadrant of a circle whose circumference is 44 cm .

Answer:

38.5cm^2

Solution:

1. Use the circumference $2\pi r = 44$ to find the radius r .
2. Calculate the area of the quadrant using $\frac{1}{4}\pi r^2$.

Derivation:

$$\begin{aligned}2 \times \frac{22}{7} \times r &= 44 \\ \Rightarrow r &= 7 \text{ cm} \\ \Rightarrow \text{Area} &= \frac{1}{4} \times \frac{22}{7} \times 7^2 \\ \Rightarrow \text{Area} &= \frac{1}{4} \times 22 \times 7 = \frac{154}{4} = 38.5 \text{ cm}^2\end{aligned}$$

Areas Related to Circles — Area of a Quadrant

- 15. 15.** The radius of two circles are 8 cm and 6 cm respectively. Find the radius of the circle having area equal to the sum of the areas of the two circles.

Answer:

$$10\text{cm}$$

Solution:

1. Set the area of the new circle πR^2 equal to $\pi r_1^2 + \pi r_2^2$.
2. Solve $R^2 = 8^2 + 6^2$ to find R .

Derivation:

$$\pi R^2 = \pi(8^2) + \pi(6^2)$$

$$\Rightarrow R^2 = 64 + 36 = 100$$

$$\Rightarrow R = \sqrt{100} = 10 \text{ cm}$$

Areas Related to Circles — Area of Combined Circles

- 16. 16.** Find the area of a sector of a circle with radius 6 cm if the angle of the sector is 60° . Use $\pi = 3.14$.

Answer:

$$18.84\text{cm}^2$$

Solution:

1. Identify the radius $r = 6 \text{ cm}$ and central angle $\theta = 60^\circ$.
2. Use the formula for the area of a sector: $A = \frac{\theta}{360^\circ} \times \pi r^2$.
3. Substitute the values and calculate the final result.

Derivation:

$$A = \frac{60}{360} \times 3.14 \times 6^2$$

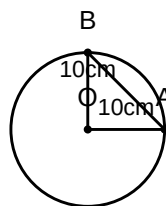
$$A = \frac{1}{6} \times 3.14 \times 36$$

$$A = 3.14 \times 6$$

$$A = 18.84 \text{ cm}^2$$

Areas Related to Circles — Area of a Sector

- 17. 17.** In a circle of radius 10 cm, a chord AB subtends a right angle at the centre O . Find the area of the minor segment. (Use $\pi = 3.14$).



Answer:

$$28.5\text{cm}^2$$

Solution:

1. Calculate the area of the sector OAB with $\theta = 90^\circ$.
2. Calculate the area of $\triangle OAB$ which is a right-angled triangle with base and height equal to 10 cm .
3. Subtract the area of the triangle from the area of the sector to get the segment area.

Derivation:

$$\text{Area of sector} = \frac{90}{360} \times 3.14 \times 10^2 = 78.5\text{ cm}^2$$

$$\text{Area of } \triangle OAB = \frac{1}{2} \times 10 \times 10 = 50\text{ cm}^2$$

$$\text{Area of segment} = 78.5 - 50 = 28.5\text{ cm}^2$$

Areas Related to Circles — Area of a Segment

18. **18.** The length of the minute hand of a clock is 14 cm . Find the area swept by the minute hand in 5 minutes.

Answer:

$$51.33\text{cm}^2$$

Solution:

1. Determine the angle swept in 60 minutes (360°), so in 5 minutes it is 30° .
2. Use the radius $r = 14\text{ cm}$.
3. Apply the sector area formula.

Derivation:

$$\theta = \frac{360}{60} \times 5 = 30^\circ$$

$$\text{Area} = \frac{30}{360} \times \frac{22}{7} \times 14 \times 14$$

$$\text{Area} = \frac{1}{12} \times 22 \times 14 = \frac{616}{12} = 51.33\text{ cm}^2$$

Areas Related to Circles — Applications of Sector Area

19. **19.** The diameters of two circles are 8 cm and 6 cm respectively. Find the radius of a circle whose area is equal to the sum of the areas of the two circles.

Answer:

$$5\text{cm}$$

Solution:

1. Find the radii of the two circles: $r_1 = 4\text{ cm}$, $r_2 = 3\text{ cm}$.
2. Equate the sum of areas $\pi r_1^2 + \pi r_2^2$ to πR^2 .
3. Solve for R .

Derivation:

$$\pi(4)^2 + \pi(3)^2 = \pi R^2$$

$$16\pi + 9\pi = \pi R^2$$

$$25\pi = \pi R^2$$

$$R^2 = 25$$

$$\Rightarrow R = 5 \text{ cm}$$

Areas Related to Circles — Area of a Circle

- 20. 20.** A horse is tethered to a peg at one corner of a square-shaped grass field of side 15 m by means of a 5 m long rope. Find the area of that part of the field in which the horse can graze.

Answer:

$$19.625m^2$$

Solution:

1. Identify the shape as a quadrant of a circle with radius $r = 5 \text{ m}$.
2. The grazing area is the area of this quadrant.
3. Calculate $\frac{1}{4} \times \pi r^2$.

Derivation:

$$\text{Area} = \frac{1}{4} \times 3.14 \times 5^2$$

$$\text{Area} = \frac{1}{4} \times 3.14 \times 25$$

$$\text{Area} = 0.785 \times 25 = 19.625 \text{ m}^2$$

Areas Related to Circles — Area of a Sector

- 21. 21.** A round table cover of radius 28 cm has a design in the middle leaving an equilateral triangle ABC in the middle as shown in the figure. Find the area of the design (shaded region). Use $\sqrt{3} = 1.73$ and $\pi = \frac{22}{7}$.

Answer:

$$\text{Area of shaded region} = 1514.88 \text{ cm}^2$$

Solution:

1. Calculate the area of the circle with $r = 28 \text{ cm}$.
2. The triangle ABC is equilateral inscribed in the circle. The side length a of the equilateral triangle is $r\sqrt{3}$.
3. Calculate the area of the equilateral triangle using $\frac{\sqrt{3}}{4}a^2$.
4. Subtract the area of the triangle from the area of the circle to get the area of the shaded design.

Derivation:

$$\text{Area}_{\text{circle}} = \pi r^2 = \frac{22}{7} \times 28 \times 28 = 2464 \text{ cm}^2$$

$$\text{Side } a = 28\sqrt{3} \text{ cm}$$

$$\text{Area}_{\Delta} = \frac{\sqrt{3}}{4}(28\sqrt{3})^2 = \frac{1.73}{4} \times 784 \times 3 = 1.73 \times 196 \times 3 = 1017.36 \text{ cm}^2$$

$$\text{Area}_{\text{shaded}} = 2464 - 1017.36 = 1446.64 \text{ cm}^2$$

- 22. 22.** A car has two wipers which do not overlap. Each wiper has a blade of length 25 cm sweeping through an angle of 115° . Find the total area cleaned at each sweep of the blades. (Take $\pi = \frac{22}{7}$)

Answer:

$$Total\ area = \frac{63250}{63} cm^2 \text{ or } approx 1003.97 cm^2$$

Solution:

1. Identify that each wiper sweeps a sector of a circle.
2. Radius $r = 25$ cm and central angle $\theta = 115^\circ$.
3. Area of one sector is $\frac{\theta}{360^\circ} \times \pi r^2$.
4. Multiply the area of one sector by 2 since there are two wipers.

Derivation:

$$\begin{aligned} Area &= 2 \times \left(\frac{115}{360} \times \frac{22}{7} \times 25 \times 25 \right) \\ Area &= 2 \times \left(\frac{23}{72} \times \frac{22}{7} \times 625 \right) \\ Area &= \frac{23 \times 11 \times 625}{36 \times 7} \\ Area &= \frac{158125}{252} \approx 627.48 \times 2 = 1254.96 cm^2 \end{aligned}$$

Areas Related to Circles — Area of Sector

- 23. 23.** In a circular park of radius 21 m, a path of uniform width 3.5 m is constructed inside the park along the boundary. Find the cost of paving the path at the rate of Rs.50 per m^2 .

Answer:

$$Cost = Rs.20625$$

Solution:

1. The path forms an annulus (ring) between two circles.
2. Outer radius $R = 21$ m.
3. Inner radius $r = R - 3.5 = 17.5$ m.
4. Area of path $= \pi(R^2 - r^2)$.
5. Multiply the area by the rate 50.

Derivation:

$$\begin{aligned} Area &= \frac{22}{7}(21^2 - 17.5^2) \\ Area &= \frac{22}{7}(21 - 17.5)(21 + 17.5) \\ Area &= \frac{22}{7}(3.5)(38.5) \\ Area &= 22 \times 0.5 \times 38.5 = 11 \times 38.5 = 423.5 m^2 \\ Cost &= 423.5 \times 50 = 21175 \end{aligned}$$

- 24. 24.** A chord AB of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments of the circle. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$)

Answer:

$$\text{Minorsegment} = 20.4375\text{cm}^2, \text{Majorsegment} = 686.0625\text{cm}^2$$

Solution:

1. Calculate the area of the minor sector with $\theta = 60^\circ$.
2. Calculate the area of $\triangle OAB$ where $OA = OB = 15$ and $\angle O = 60^\circ$ (which is equilateral).
3. Minor segment area = Area of sector - Area of $\triangle OAB$.
4. $\text{Majorsegmentarea} = \text{Areaofcircle} - \text{Areaofminorsegment}$.

Derivation:

$$\text{Area}_{\text{sector}} = \frac{60}{360} \times 3.14 \times 15^2 = \frac{1}{6} \times 3.14 \times 225 = 117.75\text{ cm}^2$$

$$\text{Area}_{\triangle} = \frac{\sqrt{3}}{4} \times 15^2 = \frac{1.73}{4} \times 225 = 97.3125\text{ cm}^2$$

$$\text{Area}_{\text{minor}} = 117.75 - 97.3125 = 20.4375\text{ cm}^2$$

$$\text{Area}_{\text{circle}} = 3.14 \times 225 = 706.5\text{ cm}^2$$

$$\text{Area}_{\text{major}} = 706.5 - 20.4375 = 686.0625\text{ cm}^2$$